

Take the following matrix:

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

Start computing its first few powers by hand: A^2 , A^3 , etc. What pattern do you see? Can you explain why this pattern shows up? This might make you curious to know if there's an efficient way to compute arbitrary powers of this matrix, A^n for any number n .

Given that two eigenvectors of this matrix are

$$\vec{v}_1 = \begin{bmatrix} 2 \\ 1 + \sqrt{5} \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 2 \\ 1 - \sqrt{5} \end{bmatrix},$$

see if you can figure out a way to compute A^n by first changing to an eigenbasis, compute the new representation of A^n in that basis, then converting back to our standard basis. What does this formula tell you?

$$D = \begin{pmatrix} a & \\ & b \end{pmatrix}$$

$$D^2 = D \cdot D = \begin{pmatrix} a & \\ & b \end{pmatrix} \cdot \begin{pmatrix} a & \\ & b \end{pmatrix} = \begin{pmatrix} a^2 & 0 \\ 0 & b^2 \end{pmatrix}$$

$$D^3 = D^2 \cdot D = \begin{pmatrix} a^3 & \\ & b^3 \end{pmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad A^2, A^3, \dots, A^n$$

$$A^2 = A \cdot A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

$$A^3 = A^2 \cdot A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$$

$$A^4 = A^3 \cdot A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$$

$$A^5 = A^4 \cdot A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 5 \\ 5 & 8 \end{pmatrix}$$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$a = \{0, 1, 1, 2, 3\}$$

$$b = \{1, 1, 2, 3, 5\}$$

$$c = \{1, 1, 2, 3, 5\}$$

$$d = \{1, 2, 3, 5, 8\}$$

$$0, 1, 1, 2, 3, 5, 8, 13, \dots$$

$$F_0, F_1, F_2$$

$$A = \begin{pmatrix} F_0 & F_1 \\ F_1 & F_2 \end{pmatrix}$$

$$A^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}$$

$$A^{n+1} = A^n \cdot A = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} F_n & F_{n-1} + F_n \\ F_{n+1} & F_n + F_{n+1} \end{pmatrix}$$

$$\Rightarrow A^{n+1} = \begin{pmatrix} F_n & F_{n+1} \\ F_{n+1} & F_{n+2} \end{pmatrix}$$

(A)

$$\det(A - \underline{\lambda} \mathbb{1}) \stackrel{!}{=} 0 \quad \begin{matrix} \lambda_1 \\ \lambda_2 \end{matrix}$$

$$\rightarrow A \cdot \vec{v} = \lambda \vec{v} \quad \leftarrow$$

$$A \cdot \vec{v} = \lambda \mathbb{1} \vec{v}$$

$$A \cdot \vec{v} - \lambda \mathbb{1} \vec{v} = 0$$

$$(\underline{A - \lambda \mathbb{1}}) \cdot \vec{v} = 0$$

$$\begin{matrix} \uparrow \\ 0 \end{matrix}$$

$$(x, y)$$

$$3 \cdot \vec{v} = (3x, 3y)$$

$$\begin{pmatrix} 3 \\ 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \end{pmatrix}$$

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

$$A - \underline{\lambda \mathbb{1}} = \begin{pmatrix} 0-\lambda & 1 \\ 1 & 1-\lambda \end{pmatrix}$$

(λ_1)

$$\begin{aligned} \det(A - \lambda \mathbb{1}) &= -\lambda \cdot (1-\lambda) - 1 = 0 \\ &= \lambda^2 - \lambda - 1 = 0 \end{aligned}$$

$$\frac{1 \pm \sqrt{5}}{2} = \begin{matrix} \lambda_1 \\ \lambda_2 \end{matrix}$$

$$\lambda_1 = \frac{1 + \sqrt{5}}{2}$$

$$\lambda_2 = \frac{1 - \sqrt{5}}{2}$$

$$(A - \lambda \mathbb{1}) \cdot \vec{v} = 0$$

(λ_1)

$$\begin{pmatrix} -\lambda_1 & 1 \\ 1 & 1-\lambda_1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$-\lambda_1 \cdot x + y = 0$$

$$y = \lambda_1 x$$

$$\begin{pmatrix} 1 \\ \lambda_1 \end{pmatrix} = \vec{v}_1 \Rightarrow \lambda_1$$

$$\vec{v}_1 = \begin{pmatrix} 1 \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1+\sqrt{5}}{2} \end{pmatrix}$$

$$\vec{v}_1' = 2\vec{v}_1 = \begin{pmatrix} 2 \\ 1+\sqrt{5} \end{pmatrix}$$

$$\lambda_2 = 0 \quad \vec{v}_2 = \begin{pmatrix} 1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1-\sqrt{5}}{2} \end{pmatrix}$$

$$\vec{v}_1 = \begin{bmatrix} 2 \\ 1+\sqrt{5} \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 2 \\ 1-\sqrt{5} \end{bmatrix},$$

$$\underline{D = P^{-1} \cdot A \cdot P} = \underline{A = P \cdot D \cdot P^{-1}}$$

$$\begin{aligned} & \underline{P = P_{B,E} = ([v_1]_E \quad [v_2]_E)} \\ & \underline{P^{-1} = P_{E,B} = \frac{1}{\det P} \cdot \text{adj}(P)} \\ & \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \right) \rightarrow \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \end{aligned}$$

$$\begin{aligned} [x]_B & \xrightarrow{P} [x]_E \xrightarrow{A} [Ax]_E \xrightarrow{P^{-1}} [Ax]_B = D \end{aligned}$$

$$P = \begin{pmatrix} 2 & 2 \\ 1+\sqrt{5} & 1-\sqrt{5} \end{pmatrix}$$

$$\begin{aligned} \det P &= 2 \cdot (1-\sqrt{5}) - 2 \cdot (1+\sqrt{5}) \\ &= 2 - 2\sqrt{5} - 2 - 2\sqrt{5} \\ &= -4\sqrt{5} \end{aligned}$$

$$P^{-1} = \frac{1}{-4\sqrt{5}} \begin{pmatrix} 1-\sqrt{5} & -2 \\ -1-\sqrt{5} & 2 \end{pmatrix}$$

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$$A \cdot P = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 1+\sqrt{5} & 1-\sqrt{5} \end{pmatrix}$$

$$= \begin{pmatrix} 1+\sqrt{5} & 1-\sqrt{5} \\ 3+\sqrt{5} & 3-\sqrt{5} \end{pmatrix}$$

$$P^{-1} \cdot (A \cdot P) = \frac{1}{-4\sqrt{5}} \begin{pmatrix} 1-\sqrt{5} & -2 \\ -1-\sqrt{5} & 2 \end{pmatrix} \cdot \begin{pmatrix} 1+\sqrt{5} & 1-\sqrt{5} \\ 3+\sqrt{5} & 3-\sqrt{5} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{-10 - 2\sqrt{5}}{-4\sqrt{5}} & 0 \\ 0 & \frac{10 - 2\sqrt{5}}{-4\sqrt{5}} \end{pmatrix}$$

$$\frac{-10 - 2\sqrt{5}}{-4\sqrt{5}} = \frac{5}{2\sqrt{5}} + \frac{1}{2} = \frac{5}{2\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} + \frac{1}{2}$$

$$= \frac{5\sqrt{5}}{2 \cdot 5} + \frac{1}{2}$$

$$\lambda_1 = \frac{1+\sqrt{5}}{2}$$

$$D = P^{-1} A P = \begin{pmatrix} \lambda_1 & \\ & \lambda_2 \end{pmatrix}$$

$$A = P \cdot D \cdot P^{-1}$$

$$A^2 = A \cdot A = P \cdot D \cdot \underbrace{P^{-1} \cdot P}_{1} \cdot D \cdot P^{-1}$$

=

$$P \cdot D \cdot 1 \cdot D \cdot P^{-1}$$

$$A^2 = P \cdot D^2 \cdot P^{-1} = \dots A^n = P$$

$$\boxed{A^n = P \cdot D^n \cdot P^{-1}}$$

$$\begin{pmatrix} \lambda_1^n & \\ & \lambda_2^n \end{pmatrix}$$

$$[X]_B = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \quad B = \{v_1, v_2\}$$

$$X = c_1 \cdot \underline{v_1} + c_2 \cdot \underline{v_2}$$

$$A \cdot X = c_1 \underbrace{A \cdot v_1}_{\lambda_1 v_1} + c_2 \underbrace{A \cdot v_2}_{\lambda_2 v_2}$$

$$[AX]_B = \begin{pmatrix} \lambda_1 \cdot c_1 \\ \lambda_2 \cdot c_2 \end{pmatrix} = D [X]_B = \begin{pmatrix} \lambda_1 & \\ & \lambda_2 \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$v_1 \perp v_2$$

$$\rightarrow \left. \begin{array}{l} \cdot \quad \overline{v_1} \cdot \overline{v_2} = 0 \\ \cdot \quad \|\overline{v_1}\| = \|\overline{v_2}\| = 1 \\ \text{ortonormal} \end{array} \right\} P^{-1} = P^T$$